

The Smirnov Property for Every Nondegenerate Bivariate Gaussian Density

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August 26, 2026

Status

Candidate full solution, likely valid, pending human review.

Source Question

Mayerhofer, *The Smirnov Property for Weighted Lebesgue Spaces*, arXiv:2101.02470, Definition 2.4, calls a density w Smirnov (for $q > 1$) when

$$F(x) + G(y) \in L^q(w), \quad F, G \in L^1(w),$$

forces $F, G \in L^q(w)$. The harmless factor $1/2$ in the paper's definition has been suppressed. Remark 2.10 asks whether this property holds for a bivariate standard normal density with nonzero correlation.

Remark 2.10. A bivariate standard normal density $w(x, y)$ with non-zero correlation has an unbounded likelihood ratio $w_X w_Y / w(x, y)$, thus does not satisfy condition [\(2.8\)](#). Furthermore, this w is also not of the form of Proposition [2.6](#), whence it is not clear whether w satisfies the Smirnov property.

We answer this question affirmatively. In fact, the result holds for every nondegenerate bivariate Gaussian law and every $1 < q < \infty$.

Proof Idea

After standardization, conditional expectation between the two Gaussian coordinates is the one-dimensional Gaussian noise operator P_ρ . If $H = F(X) + G(Y) \in L^q$, then conditioning on X and on Y gives

$$A = F + P_\rho G \in L^q(\gamma), \quad B = P_\rho F + G \in L^q(\gamma).$$

The first summand cancels from $B - P_\rho A$, leaving

$$(I - P_{\rho^2})G \in L^q(\gamma).$$

The key rigidity fact is that an $L^1(\gamma)$ function whose image under $I - P_t$ lies in $L^q(\gamma)$ must itself lie in $L^q(\gamma)$, up to its constant part. On mean-zero functions, a sufficiently high power of P_t is a strict L^q contraction. This follows by interpolating its exact L^2 spectral gap with the elementary L^1 or L^∞ bound. Thus $I - P_t$ is invertible on mean-zero L^q . An L^1 fixed-point argument rules out any residual solution outside L^q .

Gaussian Noise Operators

Let γ be standard Gaussian measure on $\mathbb{R}$. For $-1 < r < 1$, define

$$(P_r f)(x) = \int_{\mathbb{R}} f(rx + \sqrt{1-r^2}z) d\gamma(z).$$

The operators P_r are contractions on every $L^s(\gamma)$, $1 \leq s \leq \infty$, by Jensen's inequality and invariance of γ . They satisfy

$$P_r P_s = P_{rs}, \quad P_0 = \Pi,$$

where $\Pi f = \int f d\gamma$ is the projection onto constants. These identities follow directly by combining the two independent Gaussian noises.

Lemma 1 (Resolvent rigidity). *Let $0 \leq t < 1$ and $1 < q < \infty$. If $g \in L^1(\gamma)$ and $(I - P_t)g \in L^q(\gamma)$, then $g - \int g d\gamma \in L^q(\gamma)$.*

Proof. Put $T = P_t - \Pi$, considered on the full space $L^s(\gamma)$. Since $P_t \Pi = \Pi P_t = \Pi$, for every integer $m \geq 1$,

$$T^m = P_{t^m} - \Pi.$$

Consequently $\|T^m\|_{L^1 \rightarrow L^1} \leq 2$ and $\|T^m\|_{L^\infty \rightarrow L^\infty} \leq 2$.

On $L^2(\gamma)$, the normalized Hermite polynomials form an orthonormal basis, $P_s H_k = s^k H_k$, and Π removes H_0 . Hence

$$\|T^m\|_{L^2 \rightarrow L^2} = t^m.$$

For completeness, the eigenvalue identity follows immediately by applying P_s to the generating function $\exp(ux - u^2/2) = \sum_{k \geq 0} H_k(x) u^k / k!$ and integrating the independent Gaussian noise.

If $1 < q < 2$, interpolation between L^1 and L^2 gives

$$\|T^m\|_{L^q \rightarrow L^q} \leq 2^{1-\theta} t^{m\theta}, \quad \theta = 2 \left(1 - \frac{1}{q}\right) > 0.$$

If $2 < q < \infty$, interpolation between L^2 and L^∞ gives the same form of estimate with $\theta = 2/q > 0$. For $q = 2$ the displayed L^2 norm is already enough. Thus, in every case, some m satisfies $\|T^m\|_{L^q \rightarrow L^q} < 1$. Therefore $I - T^m$ is invertible on L^q by a Neumann series, and so is $I - T$, since

$$(I - T)^{-1} = (I + T + \dots + T^{m-1})(I - T^m)^{-1}.$$

Set $u = (I - P_t)g$. Then $u \in L^q(\gamma)$ and $\Pi u = 0$. Let $v = (I - T)^{-1}u \in L^q(\gamma)$. Taking expectations in $(I - T)v = u$ shows $\Pi v = 0$; hence $Tv = P_t v$ and

$$(I - P_t)v = u.$$

Now

$$d = g - \Pi g - v \in L^1(\gamma), \quad \Pi d = 0, \quad P_t d = d.$$

Iteration gives $d = P_t^m d$ for all m . As $t^m \rightarrow 0$, $P_t^m d \rightarrow \Pi d = 0$ in $L^1(\gamma)$. To see this last convergence without any hidden regularity assumption, approximate d in $L^1(\gamma)$ by a bounded continuous function ϕ . Dominated convergence gives $P_s \phi \rightarrow \Pi \phi$ in $L^1(\gamma)$ as $s \rightarrow 0$, while the L^1 contraction of P_s controls the two approximation errors uniformly in s . Thus $d = 0$, and $g - \Pi g = v \in L^q(\gamma)$. $\square$

Main Result

Theorem 1. *Let (X, Y) have a nondegenerate bivariate Gaussian distribution, and let $1 < q < \infty$. If measurable functions F and G satisfy*

$$F(X), G(Y) \in L^1, \quad F(X) + G(Y) \in L^q,$$

then $F(X), G(Y) \in L^q$. Equivalently, every nondegenerate bivariate Gaussian density has the Smirnov property for every $1 < q < \infty$.

Proof. Separate affine changes of the two coordinates preserve all the stated integrability properties. We may therefore assume that X and Y are standard normal with correlation ρ . Nondegeneracy is precisely $|\rho| < 1$.

Write $H = F(X) + G(Y)$. Conditional Jensen gives

$$A(X) := \mathbb{E}[H \mid X] \in L^q, \quad B(Y) := \mathbb{E}[H \mid Y] \in L^q.$$

The conditional laws of a standard bivariate Gaussian pair yield, as one-variable identities in $L^1(\gamma)$,

$$A = F + P_\rho G, \quad B = P_\rho F + G.$$

Since P_ρ is an L^q contraction and $P_\rho^2 = P_{\rho^2}$,

$$B - P_\rho A = (I - P_{\rho^2})G \in L^q(\gamma).$$

Lemma 1, with $t = \rho^2 < 1$, implies $G - \int G d\gamma \in L^q(\gamma)$, and constants belong to $L^q(\gamma)$. Thus $G \in L^q(\gamma)$. Finally,

$$F = A - P_\rho G \in L^q(\gamma).$$

This proves the claim. (The same proof includes the independent case $\rho = 0$.) $\square$

Consequence for the Source Integral Equations

The remaining weight hypothesis in Mayerhofer's Theorem 3.1 can be evaluated explicitly. Let $p > 1$ be the exponent in that theorem (so its Smirnov exponent is the conjugate $q = p/(p-1)$). For the standardized density

$$w_\rho(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right),$$

condition (2.3) asks that $w_X w_Y / w_\rho$ belong to $L^p(w_\rho)$. Apart from a positive constant, the integrand in its p th power is the exponential of minus one half of the quadratic form with matrix

$$M = pI - \frac{p-1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}.$$

The smaller eigenvalue of M is

$$p - \frac{p-1}{1-|\rho|} = \frac{1-p|\rho|}{1-|\rho|}.$$

Thus condition (2.3) holds exactly when

$$p|\rho| < 1, \quad \text{equivalently} \quad q > \frac{1}{1-|\rho|}.$$

Consequently, Theorem 1 supplies the previously missing Smirnov hypothesis, and Mayerhofer’s uniqueness conclusion applies in this sharp parameter range whenever the theorem’s stated assumptions on the marginals also hold. Outside this range our Smirnov result remains valid, but the source theorem cannot be invoked because its separate condition (2.3) fails.

Adversarial Checks

The proof does not assume that F or G is initially square-integrable: Hermite expansions are used only to estimate the bounded operator T^m on L^2 , followed by interpolation. The actual functions enter the spectral argument only after $(I - T)^{-1}$ has been constructed on L^q .

The possible constant obstruction is handled explicitly by II. The kernel of $I - P_t$ on $L^1(\gamma)$ consists only of constants: the final L^1 convergence argument proves this rather than importing an L^2 ergodicity claim. Negative correlations cause no sign issue because the composition used in the proof is $P_\rho^2 = P_{\rho^2}$.

Novelty Check and Limitations

The four local run indexes were searched for arXiv:2101.02470, the exact source title, the bivariate-normal question, the Smirnov property, and Gaussian/Ornstein–Uhlenbeck variants; no matching solution or attempt was found. The source context confirms that Remark 2.10 and the conclusion state a genuine open problem.

A bounded web search on August 26, 2026 used the exact phrase “Smirnov property” together with “bivariate normal”, “Gaussian”, and “Ornstein–Uhlenbeck”, as well as the functional formulation involving $F(X) + G(Y)$. It found the arXiv paper and its 2024 published version, which still describes the question as unknown, but no later resolution or this noise-operator proof. This was not a comprehensive MathSciNet or zbMATH search, so the novelty assessment remains provisional.

The theorem treats exactly two Gaussian coordinates. Extending the argument to arbitrary multivariate Gaussian vectors would require controlling a coupled system of conditional-expectation operators and is not claimed here.

Human Review Recommendation

Send to a specialist in Gaussian analysis or weighted L^p spaces. The main review points are the L^q resolvent lemma, especially its L^1 fixed-point closure, and the precise scope of the uniqueness corollary relative to all the hypotheses of the source theorem.

References

- [1] E. Mayerhofer, *The Smirnov Property for Weighted Lebesgue Spaces*, arXiv:2101.02470 (2021; revised 2024), published in *Mathematics* 12 (2024), no. 19, 3135, doi:10.3390/math12193135.